



Kempower EV Charging Infrastructure Analysis

Station Expansion Decision System — A dataset-driven analysis of real-world charging behavior across countries to guide infrastructure investment.

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INFRASTRUCTURE PLANNING

DATA-DRIVEN DECISIONS

The Infrastructure Gap Is Widening

Global EV adoption is accelerating — but charging infrastructure is not keeping pace. Without proactive planning, some regions face overload risk while others remain under-served.

Rising Demand

EV adoption is surging globally, driving unprecedented charging load.

Overload Risk

High-demand regions risk exceeding existing station capacity.

Under-Served Areas

Some regions lack sufficient infrastructure to support growing EV populations.

Our Goal

Analyze demand vs. capacity and decide where expansion is needed.



Dataset Overview



What the Data Captures

Each row represents one real-world EV charging session, capturing electrical and behavioral metrics across multiple countries.

EV Model
Audi Q4 e-tron

Most use Country
Norway

Avg Power (W)
55.64

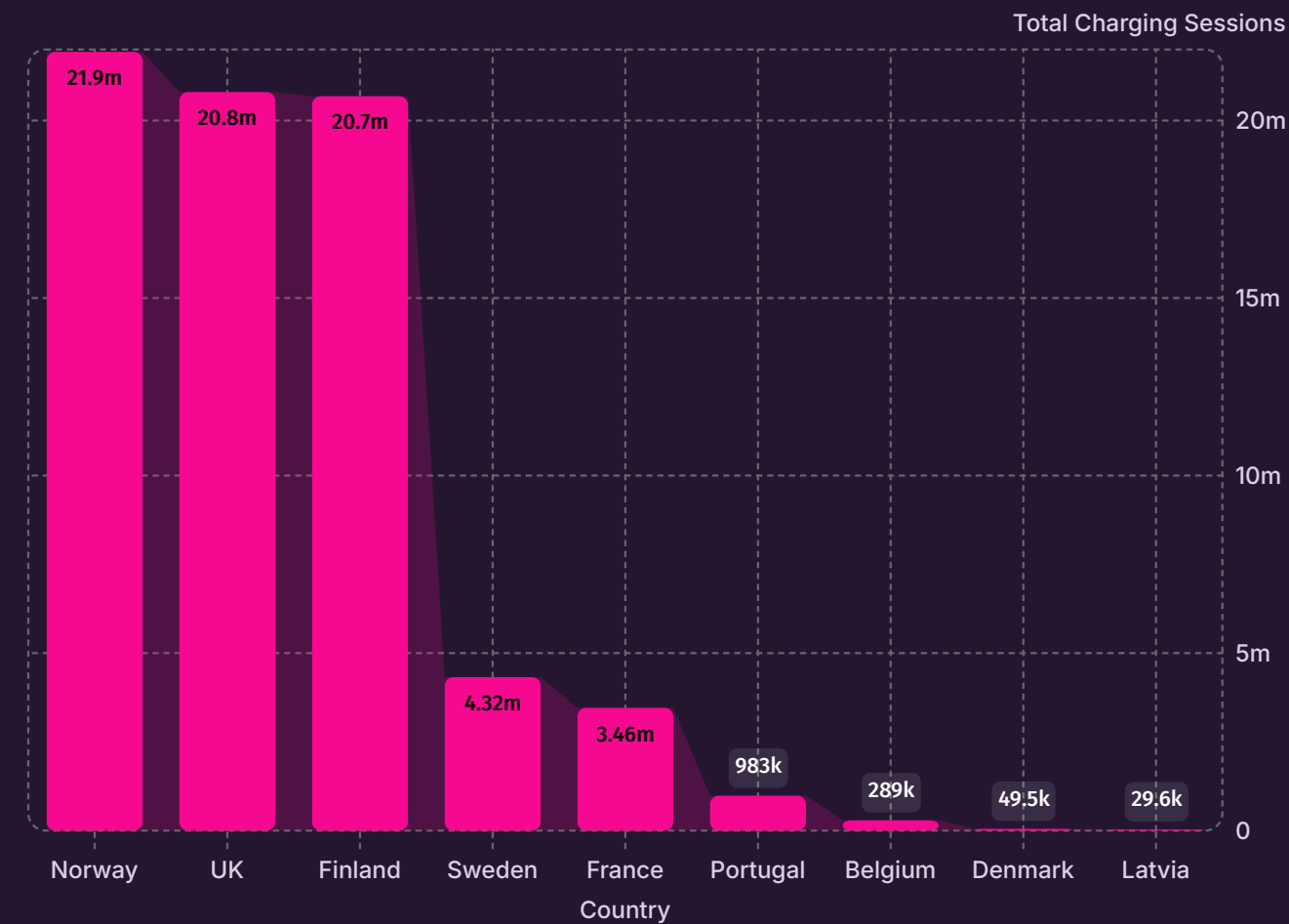
Methodology: Step-by-Step Approach

A structured five-step analytical process transforms raw session data into actionable expansion decisions.



Two core formulas drive the analysis: **Energy (kWh) = Power (kW) × Time (h)** and **Utilization Ratio = Demand ÷ Capacity**. Countries exceeding the 0.8 utilization threshold are flagged for expansion.

Data Overview: Charging Activity by Country

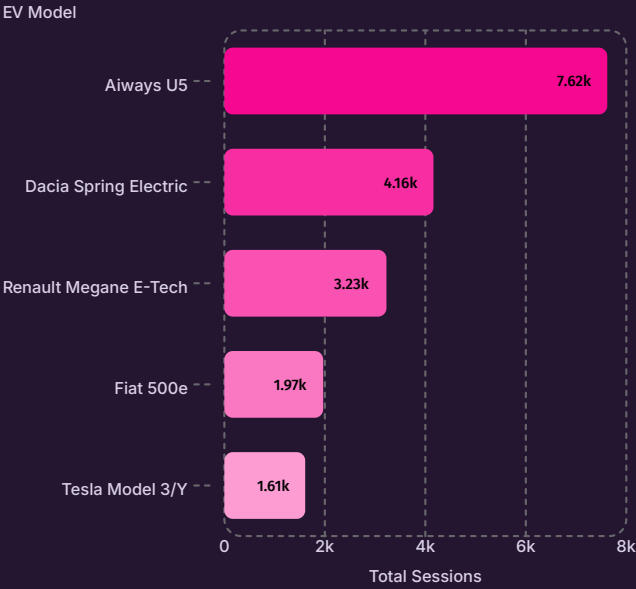


What the Data Reveals

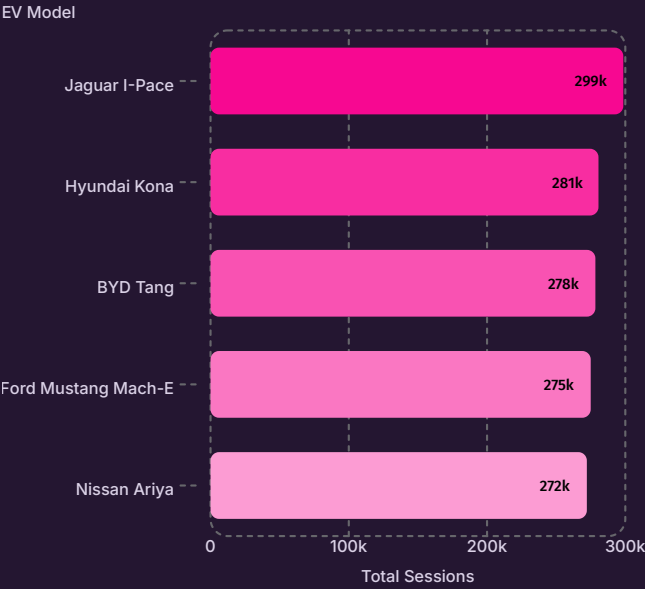
Norway and Finland lead in total charging sessions, with the United Kingdom close behind. The remaining countries show a wide range of activity levels, highlighting strong regional variation in charging demand.

Geographic Fleet Distribution: Top 5 EV Models Per Country Node

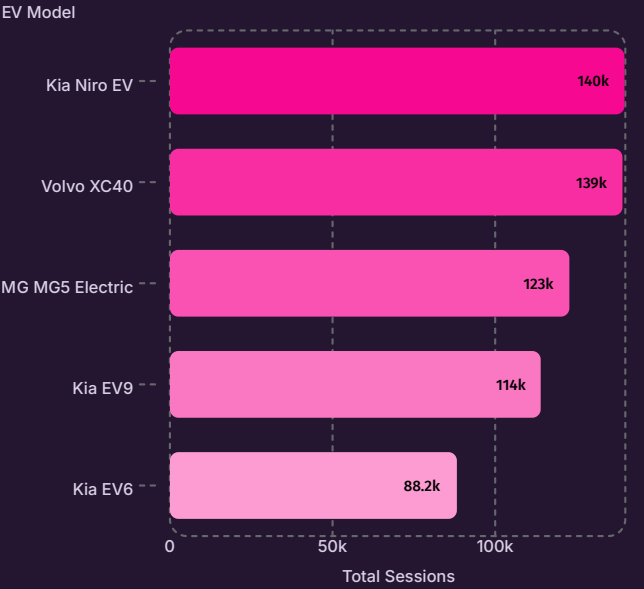
DENMARK



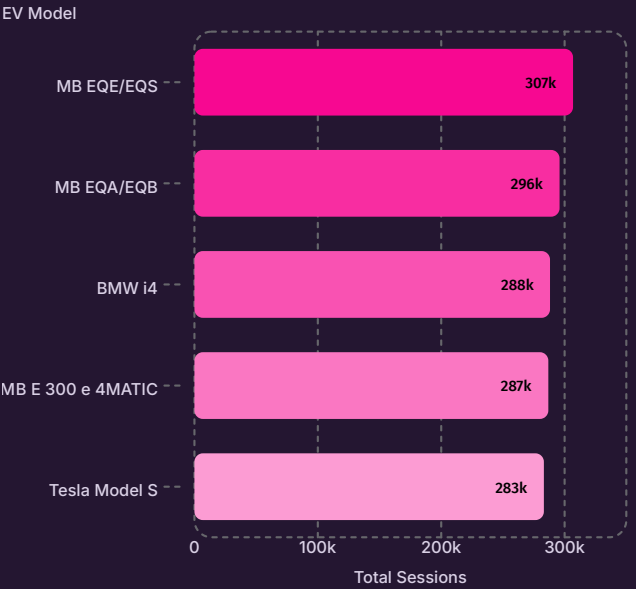
NORWAY



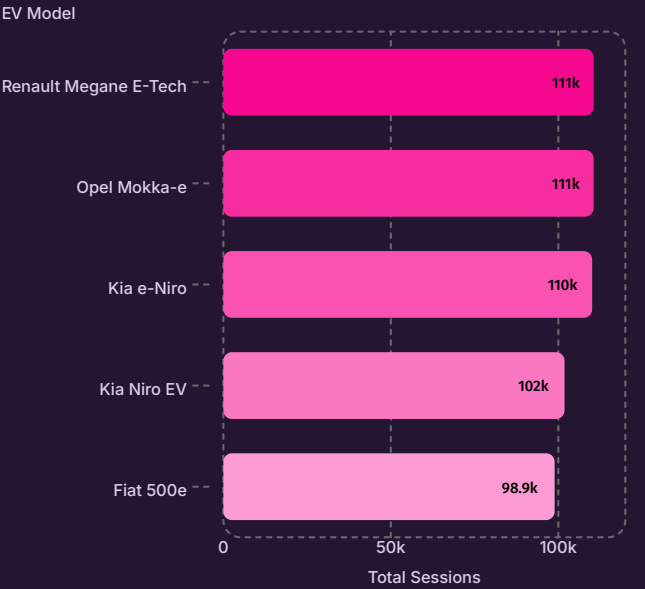
SWEDEN



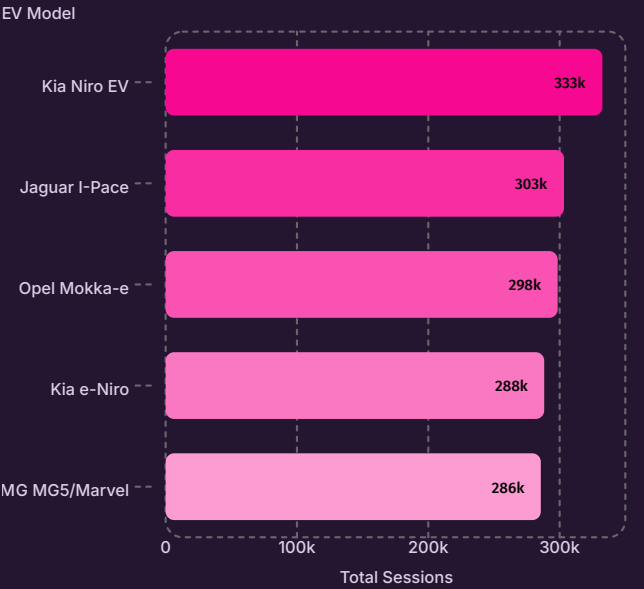
FINLAND



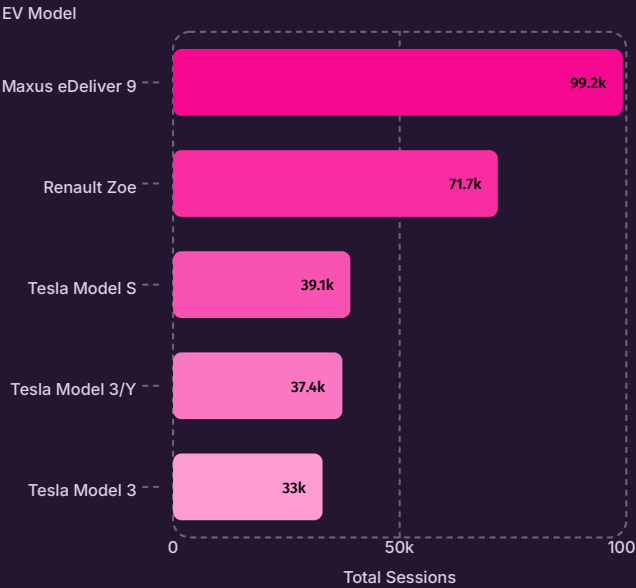
FRANCE



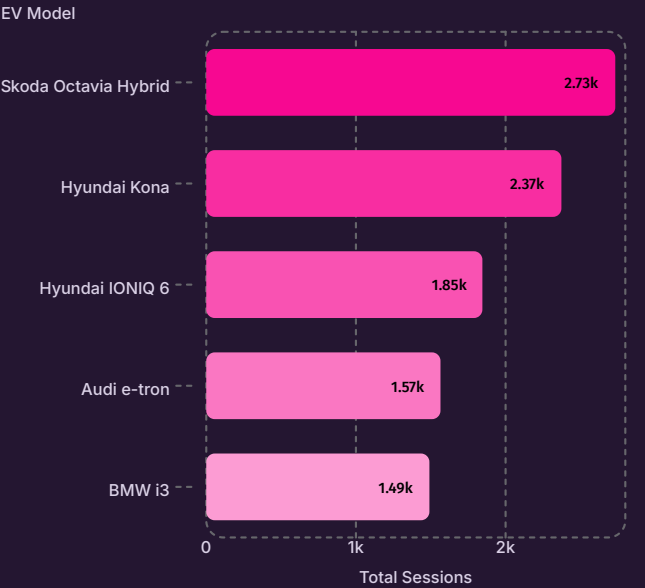
UNITED KINGDOM



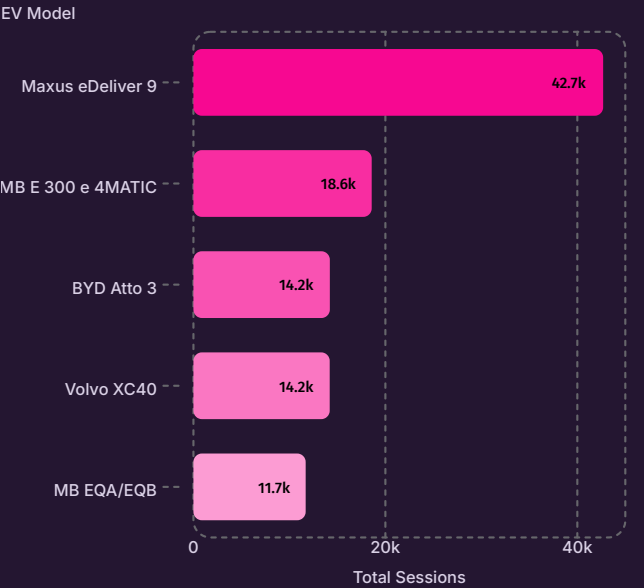
PORTUGAL



LATVIA



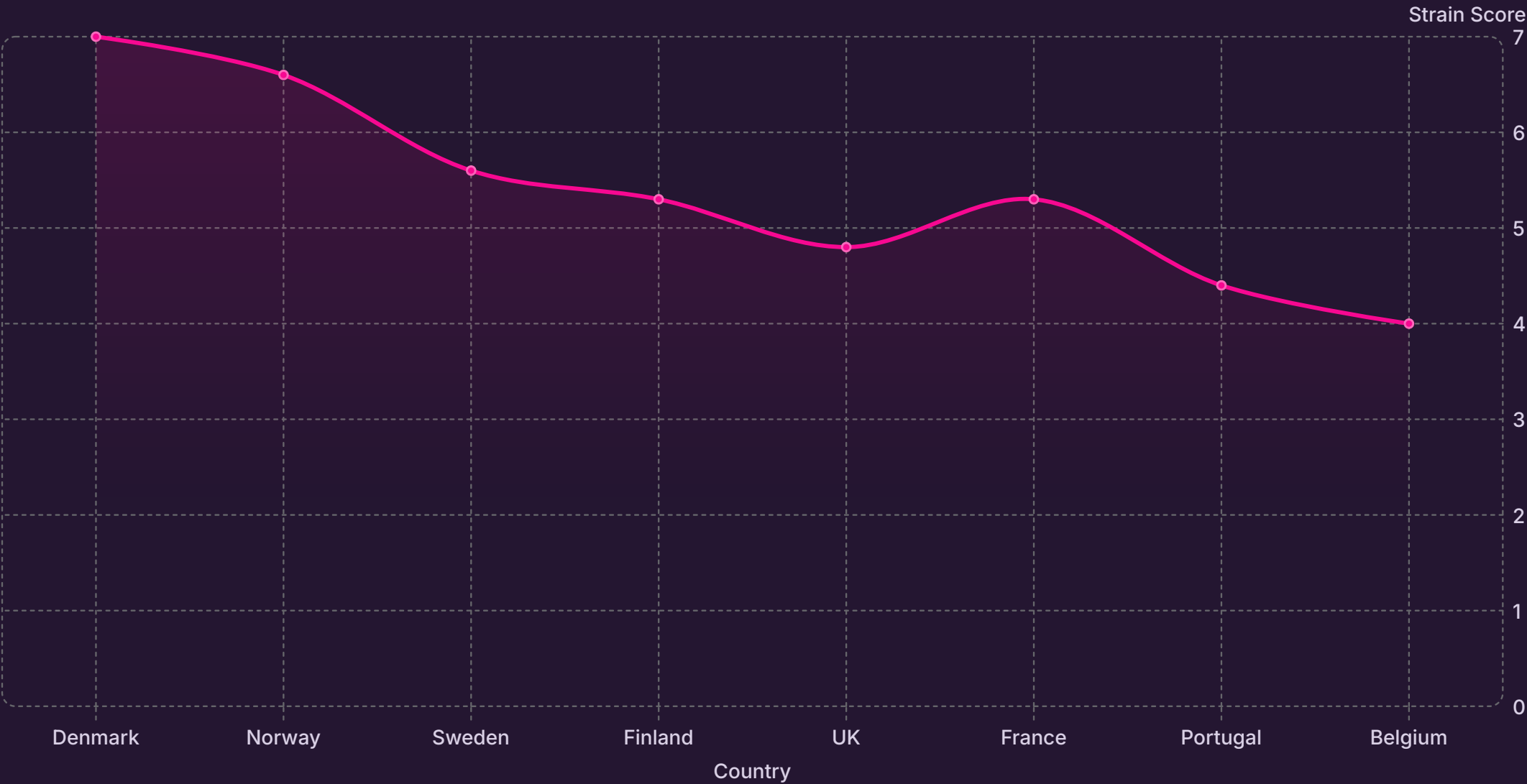
BELGIUM



Infrastructure Decision Matrix

5.38

Average Strain Score



Strain Score by Country (1-10 scale)

Country	Session Volume	2025 EV Share	Strain Score (1-10)	Strategic Decision
Denmark	49,457	71%	7.0	Warning: Expand Stalls
Norway	21,932,350	97%	6.6	Warning: Expand Stalls
Sweden	4,322,340	61%	5.6	Warning
Finland	20,682,735	57%	5.3	Stable
UK	20,800,171	35%	4.8	Stable

i Denmark exhibits the highest strain index (7.0) despite lower absolute session volume, driven by its extreme 71% EV market penetration rate—the highest globally. This reveals that market saturation, not just volume, is a critical infrastructure stress factor.

The Core Predictive Formula

$$\text{StrainScore} = (0.4 * \text{NormalizedVolume}) + (0.6 * \text{EV MarketShare})$$

Raw Session Counts (millions of data rows) and EV Market Shares (percentages) use completely different units. They are first harmonized onto a unified 1-to-10 scale using Min-Max Normalization:

$$\text{NormalizedValue} = 1 + [(X - X_{min}) / (X_{max} - X_{min})] * 9$$

Strategic Justification of Weights

Proactive vs. Reactive Sizing

Weighting Session Volume at 40% accounts for current usage, while weighting EV Share higher at 60% tracks "adoption velocity." This makes the model forward-looking instead of just highlighting currently busy spots.

Capturing 'Infrastructure Shock'

Mature markets like Norway have high volumes but stable growth curves that their grids already expect. Newer markets like Denmark have low volumes but explosive 71% EV penetration. The 60% weight ensures the model detects sudden upcoming bottlenecks before long queue lines physically form.

Mathematical Balancing Act

The 0.6 multiplier gives the macro market growth enough leverage to break through the sheer weight of millions of historical data transactions, preventing smaller high-volatility nodes from being hidden by larger countries.

Empirical Grid Telemetry & Power Infrastructure Load Analysis

High-contrast infrastructure monitoring view across 9 national nodes, combining cumulative energy throughput, peak transformer strain, and sustained delivery rates.



High-Volume Energy Heavyweights (MWh)

- Finland:** 3,576.22 MWh
- Norway:** 3,435.61 MWh
- United Kingdom:** 3,221.48 MWh

These three nodes are dominant load centers, handling continuous, heavy electrical throughput at sustained scale.



Peak Transformer Strain Factors (Absolute Grid Spikes)

- Finland:** 397.87 kW Peak
- Sweden:** 365.54 kW Peak
- Norway:** 364.81 kW Peak

Critical insight: local grid transformers must be rated for at least **400 kW** to manage extreme instantaneous load shocks without blowing fuses or causing localized blackouts.



Average Delivery Rates & Node Efficiency (kW)

- Sweden:** 62.79 kW
- Finland:** 62.25 kW
- Portugal:** 60.40 kW
- Belgium:** 42.88 kW (Lowest)

Sweden and Finland's higher rates validate the model's findings about premium German luxury EVs, including Mercedes-Benz EQE/EQS and BMW i4, with advanced thermal management.

Remaining Node Metrics

Node	Average Delivery	Peak Load	Total Energy
France	53.18 kW	253.20 kW	510.69 MWh
Portugal	60.40 kW	163.85 kW	164.98 MWh
Belgium	42.88 kW	90.47 kW	34.42 MWh
Denmark	48.91 kW	241.24 kW	6.72 MWh
Latvia	58.19 kW	236.03 kW	4.79 MWh

Conclusion & Strategic Recommendations

Dark tech interface summary of the highest-priority expansion nodes and the strategic framework for proactive infrastructure allocation.

1. Denmark - Critical (Score: 7.0)

High-risk bottleneck node. Explosive 71% growth velocity overpowers a small network footprint. Requires immediate stall deployment.

2. Norway - Critical (Score: 6.6)

Mass-market load anchor with 22 million sessions. Demands continuous capacity expansion.

3. Sweden & Finland - High Priority (Scores: 5.6 - 5.3)

Premium passenger hubs with extreme ~380 kW peaks. Requires substation upgrades (>400 kVA).

4. UK, France & Logistics Nodes - Monitor

Stable passenger volumes, but commercial cargo presence (Belgium/Portugal) requires pull-through lane layouts.

Strategic Takeaway

A one-size-fits-all infrastructure strategy will fail. By combining micro transaction telemetry, macro EV market shares, and grid physics, we have built a proactive framework. This enables data-driven capital allocation, triggering hardware expansion before capacity constraints impact the driver experience or destabilize local utility networks.